Credit Card Approval Prediction

Project Description:

The Credit Card Approval Prediction system is a machine learning based web application designed to predict whether a credit card application will be approved or not based on applicant details. The system helps financial institutions automate and streamline the credit card approval process by leveraging data-driven insights, reducing manual effort, and improving decision accuracy.

The platform takes applicant information such as demographics, income, employment details, credit history, and other relevant factors as input and uses a trained ML model to predict the likelihood of approval. The application is built using Python, Flask, and Scikit-learn, with a user-friendly web interface for seamless interaction.

This solution assists banks and NBFCs in minimizing risks, improving customer experience, and making faster, more consistent credit decisions.

Scenarios

Scenario 1: A salaried individual with a stable income and good credit history applies for a credit card. The system analyzes the details and predicts a high probability of approval. The application displays the result as "Approved" along with the approval probability.

Scenario 2: A self-employed applicant with moderate income and limited credit history applies for a credit card. The system evaluates the information and predicts the likelihood of approval as low. The application displays the result as "Not Approved".

Scenario 3: An applicant with high income but existing credit defaults applies for a credit card. The system processes the data and predicts the risk level as high, recommending rejection. The result is shown as "Not Approved" with the corresponding probability.